HW 6

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1

(a) Let $D^2 = \min_{2 \le i \le n} \|X_1 - X_i\|^2$. For $0 \le t \le 1$, define a disk $B(x, \sqrt{t}) = \{y : \|y - x\|_2 \le \sqrt{t}\}$. Then, even in the worse case, when x is at the corner of the unit square, at least one quarter of the disk lies inside the unit square. Therefore, $\text{Area}(B(x, \sqrt{t}) \cap [0,1]^2) \ge \frac{1}{4}\pi(\sqrt{t})^2 = \frac{\pi t}{4}$. Since $X_2, \cdots, X_n$ are independent and uniformly distributed in $[0,1]^2$, we have:

$$\begin{aligned} P(D^2 > t) &= P(X_i \notin B(x, \sqrt{t}) \text{ for all } i = 2, \cdots, n | X_1 = x) \\ &= [1 - \text{Area}(B(x, \sqrt{t}) \cap [0,1]^2)]^{n-1} \\ &\le \left(1 - \frac{\pi t}{4}\right)^{n-1} \end{aligned}$$

Use the fact that $1 - x \le e^{-x}$ for all $x \in \mathbb{R}$, we have $P(D^2 > t | X_1 = x) \le e^{-(n-1)\pi t/4}$. This bound doesn't actually depend on x, so $P(D^2 > t) \le e^{-(n-1)\pi t/4}$ for $0 \le t \le 1$. Using the tail integral identity for a nonnegative random variable X, we have $E[D^2] = \int_0^\infty P(D^2 > t) dt$. Since any two points in the unit square are at most $\sqrt{2}$ apart, $D^2 \le 2$, so we can bound the integral as follows: $E[D^2] = \int_0^1 P(D^2 > t) dt + \int_1^2 P(D^2 > t) dt$. We can simplify the first term:

$$\begin{aligned} \int_0^1 P(D^2 > t) dt &\le \int_0^1 e^{-(n-1)\pi t/4} dt \\ &\le \int_0^\infty e^{-(n-1)\pi t/4} dt \\ &= \frac{4}{(n-1)\pi} \end{aligned}$$

For $1 \le t \le 2$, $P(D^2 > t) \le P(D^2 > 1) \le e^{-(n-1)\pi/4}$, so we can bound the second term simply at $e^{-(n-1)\pi/4}$. Therefore, we have $E[D^2] \le \frac{4}{(n-1)\pi} + e^{-(n-1)\pi/4}$, which is $O(1/n)$, since for $n \ge 2$, $\frac{1}{n-1} \le \frac{2}{n}$ and exponential decay is faster than $1/n$. Thus, there exists some absolute constant $C > 0$ such that $E[D^2] \le C/n$ for $n \ge 2$.

(b) Let $Z = f(X_1, \cdots, X_n)$, the total length of the shortest traveling salesman path through all n points. We use the Efron-Stein inequality $Var(Z) \leq \sum_{i=1}^{n} EVar^{(i)}(Z)$. For each i, let $Z_{-i} = f(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n)$, the optimal path length after deleting X_i. Define the nearest neighbor distance of X_i among the other points as $d_i = \min_{j \neq i} \|X_i - X_j\|_2$. We first show that $0 \leq Z - Z_{-i} \leq 2d_i$. The lower bound is trivial, since removing a point cannot increase the optimal path length. For the upper bound, let X_j be the nearest neighbor of X_i so that their distance is d_i. Start with an optimal path through $n-1$ points excluding X_i, then insert X_i between X_j and X_k, its nearest neighbor. The increase in path length is consequently $\|X_j - X_i\|_2 + \|X_i - X_k\|_2 - \|X_j - X_k\|_2$. By the triangle inequality, $\|X_j - X_k\|_2 \leq \|X_j - X_i\|_2 + \|X_i - X_k\|_2$, so the increase in path length is at most $2\|X_j - X_i\|_2 = 2d_i$. Therefore, $0 \leq Z - Z_{-i} \leq 2d_i$.

For fixed X_{-i}, Z_{-i} is constant with respect to X_i. For any random variable Y and constant a, $Var(Y) = E[(Y-a)^2] - (E[Y]-a)^2 \leq E[(Y-a)^2]$. since the mean minimizes the mean squared error. Therefore, conditionally on X_{-i}, we have $Var^{(i)}(Z) \leq E^{(i)}[(Z - Z_{-i})^2] \leq E^{(i)}[4d_i^2] = 4E^{(i)}[d_i^2]$. Taking the expectation over the remaining variables, we have $EVar^{(i)}(Z) \leq 4E[d_i^2]$. By Efron-Stein, $Var(Z) \leq \sum_{i=1}^{n} EVar^{(i)}(Z) \leq 4\sum_{i=1}^{n} E[d_i^2]$. Since all X_i are iid, all d_i follow the same distribution, so $Var(Z) \leq 4nE[d_1^2]$. By part (a), $Var(Z) \leq 4nE[d_1^2] = O(1)$, since $E[d_1^2] = O(1/n)$, thus $Var(Z) \leq C$ for some absolute constant $C > 0$.

2

Let $X \sim \text{Exp}(1)$ and $g : \mathbb{R} \to \mathbb{R}$ be bounded with a bounded derivative. Define $h(x) = g(x) - g(0)$, so that $h(0) = 0$ and $h'(x) = g'(x)$. Since variance is unchanged by subtracting a constant, $Var(g(X)) = Var(h(X))$, then $Var(h(X)) = E[h(X)^2] - (E[h(X)])^2 \leq E[h(X)^2]$. Now, using the hint and since $(h(x)^2)' = 2h(x)h'(x)$, we have:

$$\begin{aligned} E[h(X)^2] &= \int_0^\infty h(x)^2 e^{-x} dx \\ &= h(0)^2 + \int_0^\infty 2h(x)h'(x)e^{-x} dx \\ &= 2E[h(X)h'(X)] \end{aligned}$$

Then, by Cauchy-Schwarz

$$\begin{aligned}
E[h(X)h'(X)] &\leq 2\sqrt{E[h(X)^2]E[h'(X)^2]} \\
\sqrt{E[h(X)^2]} &\leq 2\sqrt{E[h'(X)^2]} \\
E[h(X)^2] &\leq 4E[h'(X)^2] \\
Var(g(X)) &= Var(h(X)) \\
&\leq E[h(X)^2] \\
&\leq 4E[h'(X)^2] \\
&= 4E[g'(X)^2]
\end{aligned}$$

Thus, the one dimensional Poincare inequality is $Var(g(X)) \leq 4E[g'(X)^2]$. Now, we expand to n dimensions using the Efron-Stine inequality. Let $Z = f(X_1, \ldots, X_n)$. Fix $i \in \{1, \ldots, n\}$ and let $g_i(x) = f(X_1, \ldots, X_{i-1}, x, X_{i+1}, \ldots, X_n)$. Because f and its first partial derivatives are bounded, g_i is bounded with a bounded derivative. $g_i'(x) = \frac{\partial f}{\partial x_i}(X_1, \ldots, X_{i-1}, x, X_{i+1}, \ldots, X_n)$. Since X_i is independent of X_{-i}, its conditional distribution given X_{-i} remains Exp(1). Therefore, we can apply the one dimensional Poincare inequality:

$$\begin{aligned}
Var^{(i)}(f(X_1, \ldots, X_n)) &= Var(g_i(X_i)|X_{-i}) \\
&\leq 4E[g_i'(X_i)^2|X_{-i}] \\
&= 4E\left[\left(\frac{\partial f}{\partial x_i}(X_1, \ldots, X_n)\right)^2 |X_{-i}\right] \\
E[Var^{(i)}(Z)] &\leq 4E\left[E\left[\left(\frac{\partial f}{\partial x_i}(X_1, \ldots, X_n)\right)^2 |X_{-i}\right]\right] \\
&= 4E\left[\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right]
\end{aligned}$$

Now, use Efron-Stein:

$$\begin{aligned}
Var(f(X)) &\leq \sum_{i=1}^{n} E[Var^{(i)}(Z)] \\
&\leq 4\sum_{i=1}^{n} E\left[\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right] \\
&= 4E\left[\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right]
\end{aligned}$$

By the definition of the squared Euclidean norm of the gradient, $\|\nabla f(X)\|_2^2 = \sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}(X)\right)^2$, thus $Var f(X_1, \ldots, X_n) \leq 4E[\|\nabla f(X_1, \ldots, X_n)\|_2^2]$, which is the n-dimensional Poincare inequality.

3

(a)

Let $Z = X_{(k)}$. For each $i \in \{1, \dots, n\}$, define Z_{-i} to be the k-th largest observation among the $n-1$ values obtained by removing X_i. Since $k \leq n-1$, Z_{-i} is well defined. By the Efron-Stein inequality, we know that $Var(Z) \leq \sum_{i=1}^{n} E[Var^{(i)}(Z)]$, where $Var^{(i)}(Z) = Var(Z|X_{-i})$ and X_{-i} denotes all observations except X_i. Conditionally on X_{-i}, Z_{-i} is a constant because it does not depend on X_{-i}. We know that for any random variable Y and constant a:

$$\begin{aligned} E[(Y-a)^2] &= E[(Y - EY + EY - a)^2] \\ &= Var(Y) + (EY - a)^2 \\ &\geq Var(Y) \end{aligned}$$

Applying this conditionally with $Y = Z$ and $a = Z_{-i}$, we get $Var^{(i)}(Z) \leq E[(Z - Z_{-i})^2 | X_{-i}]$, and thus $Var(Z) \leq \sum_{i=1}^{n} E[(Z - Z_{-i})^2]$.

Let $\Delta = X_{(k)} - X_{(k+1)}$. Because observations are ordered in decreasing order, we know that $\Delta \geq 0$. We need to consider two cases:
Case 1: X_i is among the k largest observations
Then removing X_i makes the new largest kth value become the original $(k+1)$-th largest value, so $Z_{-i} = k+1$, and therefore $Z - Z_{-i} = X_{(k)} - X_{(k+1)} = \Delta$.
Case 2: X_i is not among the k largest observations
Then removing X_i does nothing to the k largest observations and $Z - Z_{-i} = 0$.

Now we sum over all observations with respect to the two cases:

$$\begin{aligned} \sum_{i=1}^{n} (Z - Z_{-i})^2 &= \underbrace{\Delta^2 + \cdots + \Delta^2}_{k \text{ terms}} + \underbrace{0 + \cdots + 0}_{n-k \text{ terms}} \\ &= k\Delta^2 \\ &= k(X_{(k)} - X_{(k+1)})^2 \\ \sum_{i=1}^{n} E[(Z - Z_{-i})^2] &= E[\sum_{i=1}^{n} (Z - Z_{-i})^2] \\ &= kE[(X_{(k)} - X_{(k+1)})^2] \end{aligned}$$

Then, by Efron-Stein, we finally get:

$$\begin{aligned} Var(X_{(k)}) &= Var(Z) \\ &\leq \sum_{i=1}^{n} E[(Z - Z_{-i})^2] \\ &= kE[(X_{(k)} - X_{(k+1)})^2] \end{aligned}$$

Thus, $Var(X_{(k)}) \leq kE[(X_{(k)} - X_{(k+1)})^2]$ for any $1 \leq k \leq n-1$.

(b)

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

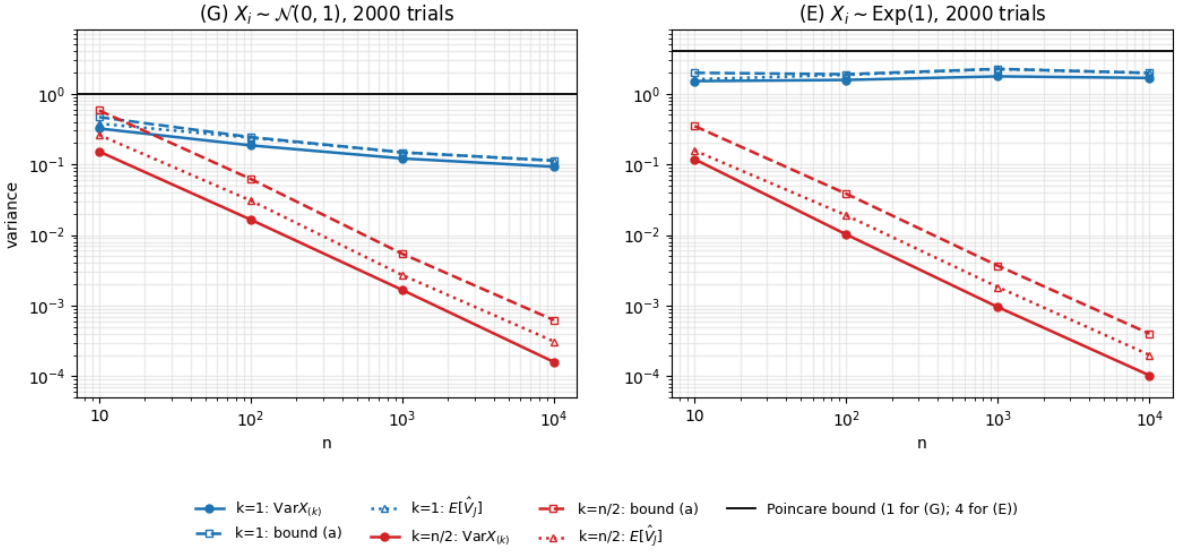
TRIALS = 2000
SIZES = [10, 100, 1000, 10000]
BATCH = 200
rng = np.random.default_rng(0)
rows = []

for dist in ('Gaussian', 'Exponential'):
    for n in SIZES:
        measurements = {kname: {'value': [], 'gap2': [], 'jackknife': []}
                        for kname in ('maximum', 'median')}
        for start in range(0, TRIALS, BATCH):
            count = min(BATCH, TRIALS - start)
            if dist == 'Gaussian':
                x = rng.standard_normal((count, n))
            else:
                x = rng.exponential(scale=1.0, size=(count, n))
            for kname, k in (('maximum', 1), ('median', n // 2)):
                p = np.partition(x, (n-k-1, n-k), axis=1)
                xk = p[:, n-k]
                xk1 = p[:, n-k-1]
                gap2 = (xk - xk1)**2
                jackknife = ((n-1)/n) * k * (1-k/n) * gap2
                measurements[kname]['value'].extend(xk.tolist())
                measurements[kname]['gap2'].extend(gap2.tolist())
                measurements[kname]['jackknife'].extend(jackknife.tolist())
        for kname, k in (('maximum', 1), ('median', n // 2)):
            values = np.asarray(measurements[kname]['value'])
            gap2 = np.asarray(measurements[kname]['gap2'])
            jack = np.asarray(measurements[kname]['jackknife'])
            v = values.var(ddof=1)
            b = k * gap2.mean()
            j = jack.mean()
            rows.append({'distribution': dist, 'n': n, 'k': k,
                         'statistic': kname, 'variance': v, 'bound_a': b,
                         'mean_jackknife': j,
                         'relative_sd_jackknife': jack.std(ddof=1)/j,
                         'bound_to_variance': b/v,
                         'jackknife_to_variance': j/v})
```

```
df = pd.DataFrame(rows)
```

In [11]:
```
fig, axes = plt.subplots(1, 2, figsize=(12.5, 5.0), sharex=True)
colors = {'maximum': 'tab:blue', 'median': 'tab:red'}
style = [('variance', '-', 'o', r'Var$X_{(k)}$'), ('bound_a', '--', 's', 'bound (a)
         ('mean_jackknife', ':', '^', r'$E[\hat{V}_J]$')]
for ax, dist in zip(axes, ('Gaussian', 'Exponential')):
    sub = df[df.distribution == dist]
    for kname in ('maximum', 'median'):
        data = sub[sub.statistic == kname]
        for col, ls, marker, label in style:
            ax.plot(data.n, data[col], ls, marker=marker, linewidth=1.8,
                    markersize=5, color=colors[kname],
                    label=f"k={'1' if kname == 'maximum' else 'n/2'}: {label}",
                    markerfacecolor='none' if col != 'variance' else colors[kname])
    pc = 1 if dist == 'Gaussian' else 4
    ax.axhline(pc, color='black', linewidth=1.3, linestyle='-',
               label='Poincare bound (1 for (G); 4 for (E))')
    ax.set_xscale('log')
    ax.set_yscale('log')
    ax.set_xticks(SIZES, ['10', r'10$^2$', r'10$^3$', r'10$^4$'])
    if dist == 'Gaussian':
        ax.set_title(r'(G) $X_i \sim \mathcal{N}(0, 1)$' + f', {TRIALS} trials')
    else:
        ax.set_title(r'(E) $X_i \sim \mathrm{Exp}(1)$' + f', {TRIALS} trials')
    ax.set_xlabel('n')
    ax.grid(True, which='both', alpha=0.22)
    ax.set_ylim(5e-5, 8)
axes[0].set_ylabel('variance')
handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, labels, loc='lower center', bbox_to_anchor=(0.5, -0.06),
           ncol=4, fontsize=9, frameon=False)
fig.subplots_adjust(bottom=0.23, wspace=0.19)
plt.show()

print(df.to_string(index=False, float_format=lambda z: f'{z:.6g}'))
```



distribution	n	k	statistic	variance	bound_a	mean_jackknife	relative_sd_jackknife	bound_to_variance	jackknife_to_variance
Gaussian	10	1	maximum	0.320902	0.464946	0.376606	1.65096	1.44887	1.17359
Gaussian	10	5	median	0.151616	0.57883	0.260473	1.8184	3.81774	1.71798
Gaussian	100	1	maximum	0.185226	0.242309	0.237487	1.97451	1.30818	1.28215
Gaussian	100	50	median	0.01647	0.0619844	0.0306823	2.07567	3.76348	1.86292
Gaussian	1000	1	maximum	0.120886	0.147417	0.147122	1.92604	1.21947	1.21703
Gaussian	1000	500	median	0.00166544	0.00541575	0.00270517	2.39296	3.25185	1.6243
Gaussian	10000	1	maximum	0.0924339	0.112739	0.112717	1.99764	1.21967	1.21943
Gaussian	10000	5000	median	0.000159376	0.000623341	0.000311639	2.10235	3.91114	1.95538
Exponential	10	1	maximum	1.50107	1.96738	1.59358	2.30665	1.31065	1.06163
Exponential	10	5	median	0.118025	0.347289	0.15628	2.05495	2.9425	1.32412
Exponential	100	1	maximum	1.56115	1.88037	1.84295	2.15488	1.20448	1.18051
Exponential	100	50	median	0.0101714	0.0383104	0.0189637	2.08279	3.7665	1.86442
Exponential	1000	1	maximum	1.7531	2.23194	2.22748	2.60704	1.27314	1.27059
Exponential	1000	500	median	0.000954521	0.00367163	0.00183398	2.05535	3.84657	1.92136
Exponential	10000	1	maximum	1.66509	1.95843	1.95804	2.16832	1.17617	1.17594
Exponential	10000	5000	median	0.000103092	0.000401382	0.000200671	2.28592	3.89344	1.94652

i.

Gaussian max: Variance decreases slowly as n increases at a rate of $\frac{1}{\log n}$. Both bound (a) and the expected jackknife follow the same order of decrease. Poincare bound remains constant at 1.

Gaussian median: Variance decreases slowly as n increases at a rate of around $\frac{1}{n}$. Both bound (a) an the expected jackknife follow the same order of decrease but overestimate the variance.

Exponential max: Variance approaches a positive constant rather than decreasing to zero. Bound (a) and the expected jackknife approach 2, so they have the correct constant-order dependence.

Exponential median: Variance decreases slowly as n increases at a rate of around $\frac{1}{n}$. Both bound (a) an the expected jackknife follow the same order of decrease.

The Poincare bound doesn't capture these rates because for the k-th largest order statistic, $f(x) = x_{(k)}$. Whenever all coordinates are distinct, exactly one coordinate determines the k-th largest value. Hence, the gradient only has one entry equal to 1 and all other entries are equal to 0. Therefore, $\|\nabla f(x)\|_2^2 = 1$ almost everywhere, giving $E[\|\nabla f(X)\|_2^2] = 1$, and

any upper bound of the form $CE[\nabla f(X)\|_2^2] = C$ is constant in n. Consequently, it can't capture the decreasing value for either median or max Gaussian.

ii. At $n = 10,000$: | Distribution | k | Bound (a) factor | Jackknife factor |-|-|-|-| | Gaussian | 1 | 1.22x | 1.22x | | Gaussian | 5,000 | 3.91x | 1.96x | | Exponential | 1 | 1.18x | 1.18x | | Exponential | 5,000 | 3.89x | 1.95x |

We can see that bound (a) overestimates by around 20% for $k = 1$ and 4 times for $k = 5,000$ while jackknife overestimates by around also 20% for $k = 1$ but only 2 times for $k = 5,000$.

iii. No, the simulation indicates that the ratio goes to around 2. For exponential order statistics, the adjacent spacing satisfies $\Delta = X_{(k)} - X_{(k+1)} \sim \text{Exp}(k)$. We already know that the jackknife is a constant multiple of Δ^2, so its relative standard deviation also equals that of Δ^2. For an exponential random variable with rate k, $E[\Delta^2] = \frac{2}{k^2}$, $E[\Delta^4] = \frac{24}{k^4}$. Consequently, $Var(\Delta^2) = E[\Delta^4] - (E[\Delta^2])^2 = \frac{24}{k^4} - \frac{4}{k^4} = \frac{20}{k^4}$. Therefore, the ratio is now equal to $\frac{\sqrt{20/k^4}}{2/k^2} = \sqrt{5} \sim 2.2$. Thus, for exponential observations, the jackknife's relative standard deviation is exactly $\sqrt{5}$ for every sample size and every eligible k. The Gaussian results are also consistent with this, because the appropriately rescaled adjacent order-statistic gaps approach exponential distributions.

No, we should not use the jackknife to estimate the variance of a sample median since it is unreliable for estimating variance. Its expected value is approximately twice the actual median variance in the simulation, so it has substantial upward bias. Second, its relative standard deviation does not approach zero. Even with a very large sample, one observed adjacent spacing can produce a highly variable variance estimate.

3

Let $Z = X - EX$, then $EZ = 0$, and since Z is $\frac{\sigma^2}{4}$-subgaussian, we have $E[e^{\lambda Z}] \leq e^{\frac{\lambda^2\sigma^2}{8}}$ for all $\lambda \in \mathbb{R}$. Define the log MGF as $\psi(\lambda) = \log E[e^{\lambda Z}]$. Then, $\psi(0) = 0$ and $\psi(\lambda) \leq \frac{\lambda^2\sigma^2}{8}$. Now use Jensen's inequality to get that $E[e^{\lambda Z}] \geq e^{\lambda EZ} = 1$, so $\psi(\lambda) \geq 0$. Combining these inequalities gives us $0 \leq \psi(\lambda) \leq \frac{\lambda^2\sigma^2}{8}$ for all $\lambda \in \mathbb{R}$, which also guarantees that the MGF is finite for every real λ and ψ is differentiable.

Since $\log(e^{\lambda Z}) = \lambda Z$, we can use the provided entropy function to get $\text{Ent}(e^{\lambda Z}) = \lambda E[Ze^{\lambda Z}] - E[e^{\lambda Z}] \log E[e^{\lambda Z}]$. Now, from ψ, we know $E[e^{\lambda Z}] = e^{\psi(\lambda)}$ and $E[Ze^{\lambda Z}] = \psi'(\lambda)e^{\psi(\lambda)}$, so we can rewrite the entropy as $\text{Ent}(e^{\lambda Z}) = \lambda\psi'(\lambda)e^{\psi(\lambda)} - e^{\psi(\lambda)}\psi(\lambda) = E[e^{\lambda Z}](\lambda\psi'(\lambda) - \psi(\lambda))$. Thus, it suffices to prove that $\lambda\psi'(\lambda) - \psi(\lambda) \leq \frac{\lambda^2\sigma^2}{8}$ for all $\lambda \in \mathbb{R}$.

Define the exponentially tilted probability distribution $\frac{dP_\lambda}{dP} = \frac{e^{\lambda Z}}{E[e^{\lambda Z}]}$. Differentiating twice gives us $\psi''(\lambda) = \frac{E[Z^2e^{\lambda Z}]}{E[e^{\lambda Z}]} - \left(\frac{E[Ze^{\lambda Z}]}{E[e^{\lambda Z}]}\right)^2 = E_\lambda[Z^2] - (E_\lambda[Z])^2 = Var_\lambda(Z)$, where E_λ and Var_λ are the expectation and variance under the tilted distribution P_λ, which ultimately $Var_\lambda(Z) \geq 0$, and thus the log MFG is convex. For any differentiable convex function, $\psi(b) \geq \psi(a) + \psi'(a)(b-a)$. Choose $a = \lambda$, $b = 2\lambda$, then $\psi(2\lambda) \geq \psi(\lambda) + \psi'(\lambda)\lambda$, which we can simplify:

$$\begin{aligned}
\psi(2\lambda) &\geq \psi(\lambda) + \psi'(\lambda)\lambda \\
\lambda\psi'(\lambda) &\leq \psi(2\lambda) - \psi(\lambda) \\
\lambda\psi'(\lambda) - \psi(\lambda) &\leq \psi(2\lambda) - 2\psi(\lambda) \leq \psi(2\lambda) \\
\lambda\psi'(\lambda) - \psi(\lambda) &\leq \frac{(2\lambda)^2\sigma^2}{8} = \frac{\lambda^2\sigma^2}{2}
\end{aligned}$$

Now apply this to the entropy function to get $\text{Ent}(e^{\lambda Z}) \leq \frac{\lambda^2\sigma^2}{2}E[e^{\lambda Z}]$. Recall since $X = Z + EX$, we have $e^{\lambda X} = e^{\lambda EX}e^{\lambda Z}$. For any positive constant c and positive random variable Y:

$$\begin{aligned}
\text{Ent}(cY) &= E[cY\log(cY)] - E[cY]\log E[cY] \\
&= cE[Y\log(cY)] - cE[Y]\log(cE[Y]) \\
&= cE[Y(\log c + \log Y)] - cE[Y](\log c + \log E[Y]) \\
&= cE[Y\log Y] + cE[Y]\log c - cE[Y]\log E[Y] - cE[Y]\log c \\
&= c(E[Y\log Y] - E[Y]\log E[Y]) \\
&= c\text{Ent}(Y)
\end{aligned}$$

Now apply this with $c = e^{\lambda EX}$:

$$\begin{aligned}
\text{Ent}(e^{\lambda X}) &= \text{Ent}(e^{\lambda EX} e^{\lambda Z}) \\
&= e^{\lambda EX} \text{Ent}(e^{\lambda Z}) \\
&\leq e^{\lambda EX} \frac{\lambda^2 \sigma^2}{2} E[e^{\lambda Z}] \\
&= \frac{\lambda^2 \sigma^2}{2} E[e^{\lambda EX} e^{\lambda Z}] \\
&= \frac{\lambda^2 \sigma^2}{2} E[e^{\lambda X}]
\end{aligned}$$

Thus, $\text{Ent}(e^{\lambda X}) \leq \frac{\lambda^2 \sigma^2}{2} E[e^{\lambda X}]$ for all $\lambda \in \mathbb{R}$.

AI Usage: All homework problems were written in VSCode and compiled using LaTeX (Problem 3 utilized Jupyter notebook), with assistance from ChatGPT and Github Copilot inline suggestions.